

Amplitude-Robust Qubit Spectroscopy Using Echo-Root-Lorentzian Pulse Shapes

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We present a qubit spectroscopy method based on Lorentzian-derived drive pulses that maintains narrow spectral features at drive amplitudes exceeding those used in conventional spectroscopy by more than three orders of magnitude, thereby enhancing robustness against amplitude variations. Specifically, we employ a root-Lorentzian envelope with a π -phase inversion at its peak, which introduces an echo-like symmetry that cancels on-resonant Rabi oscillations while preserving the spectral response. This approach suppresses conventional power broadening and enables high-resolution spectroscopy using pulses shorter than durations required by standard techniques. We experimentally demonstrate this method on a superconducting qubit platform and observe qualitative agreement with numerical simulations, highlighting its potential for rapid and calibration-robust qubit frequency characterization.

Accurate frequency characterization of a quantum two-level system (qubit) is essential for calibrating high-fidelity control in fault-tolerant quantum processors. In many physical implementations, such as superconducting circuits and semiconductor quantum dots, the qubit is realized within an inherently multilevel structure with discrete energy levels, allowing two levels to be selected to form an effective computational subspace that mimics atomic systems via their quantized energy spectrum [1, 2]. Spectroscopic identification of the transition frequency is therefore routinely performed by applying resonant control fields and monitoring the resulting population dynamics [3–5]. Spectroscopic resolution in direct population spectroscopy of coherently driven two-level systems is constrained by decoherence. In the absence of technical noise, stochastic phase fluctuations induce homogeneous broadening of the transition frequency, leading to a minimum resolvable linewidth that scales inversely with the coherence time T_2 , as described by the Bloch equations [6, 7]. This establishes the so-called T_2 limit for conventional direct spectroscopy.

In practice, however, driven spectroscopy introduces an additional trade-off between signal robustness and spectral resolution. When finite-amplitude control pulses are applied, the system dynamics deviate from the idealized two-level response due to coherent Rabi oscillations and coupling to noncomputational levels. Increasing the drive amplitude improves signal contrast and robustness against state-preparation and measurement errors, but simultaneously induces power broadening, resulting in a linewidth that scales with the instantaneous Rabi frequency. Consequently, conventional square-pulse and continuous-wave spectroscopic techniques must operate at low drive amplitudes to maintain resolution approaching the decoherence-limited linewidth set by T_2 , creating a severe practical trade-off between precision and robustness. Smooth temporal envelopes can weaken this broadening, and powers-of-Lorentzian pulses have been shown theoretically and experimentally to exhibit power narrowing [8, 9].

Ramsey interferometry provides a standard alternative to direct spectroscopy and, in principle, enables frequency estimation at the T_2 limit. Achieving such resolution, however, requires long interrogation times and sufficiently dense temporal sampling to resolve the resulting interference fringes. Moreover, Ramsey measurements are susceptible to systematic biases: asymmetries in the exponential decay envelope—arising, for example, from pulse miscalibration—can shift the apparent resonance peak in the frequency domain. Additional effects such as phase fluctuations or unintended AC Stark shifts during the control pulses introduce further distortions, ultimately biasing the extracted transition frequency [10, 11].

Alternative protocols can evade a probe- T_2 resolution bound by using inter-shot phase correlations, external clock references, dynamical decoupling, or tailored filter functions [12–17]. These approaches address different sensing settings and typically require repeated correlations, extended interrogation times, a stable external reference, or additional control resources. Here, we address the resolution-robustness trade-off within direct driven population spectroscopy by employing adiabatically engineered Lorentzian-derived drive pulses with a π -phase inversion at their peak. Unlike prior Lorentzian-pulse power-narrowing demonstrations, this echo-like structure suppresses on-resonant Rabi oscillations while preserving the off-resonant spectral response, maintaining narrow frequency features at drive amplitudes exceeding those used in conventional spectroscopy by more than three orders of magnitude.

Precise determination of the qubit transition frequency is crucial for implementing high-fidelity quantum gates. The drive frequency must coincide with the qubit resonance, as detuning directly reduces gate fidelity. A rigorous metric for quantifying such calibration errors is the diamond norm, which captures deviations in the implemented quantum operation from the ideal one. Minimizing these detuning-induced errors is essential to meet the fault-tolerance thresholds required for scalable quantum computation. [18]

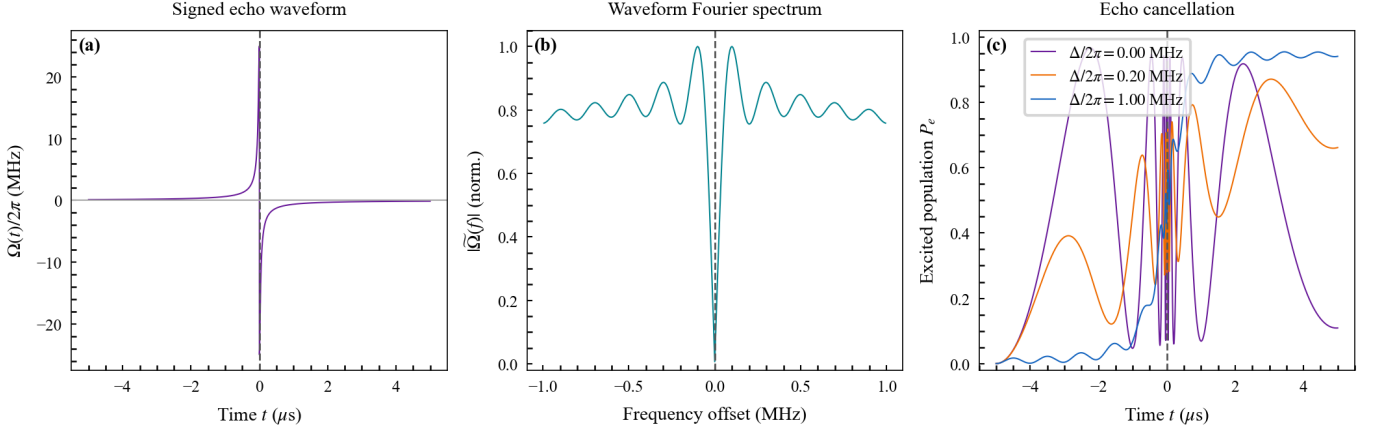


Figure 1. Pulse waveform and echo-cancellation mechanism. (a) Signed echo-root-Lorentzian drive envelope with a π phase inversion at the pulse midpoint. (b) Normalized magnitude of the waveform Fourier spectrum, showing the spectral null at zero frequency produced by the phase inversion. (c) Simulated excited-state population during the pulse at resonance and at non-resonant detunings of $\Delta/2\pi = 0.20$ and 1.00 MHz, using $T_1 = 9999 \mu\text{s}$. At resonance, the evolution accumulated during the first half is reversed during the second half, whereas finite detuning prevents complete cancellation.

Here, we propose a spectroscopic method for probing the qubit transition frequency using arbitrarily shaped drive pulses, specifically an echo-root-Lorentzian pulse. Previous theoretical work showed that Lorentzian pulses exhibit power narrowing, complementary to the conventional power broadening observed for constant drives, as a consequence of their adiabatic properties [8]. Power narrowing with powers-of-Lorentzian pulses was subsequently demonstrated experimentally on IBM Quantum, including the cutoff broadening introduced by truncating the pulse wings [9]. In this work, we extend this framework to longer pulse durations approaching the T_1 timescale and introduce a π -phase jump at the pulse peak. The resulting echo property effectively cancels the net pulse area while preserving the adiabatic properties of the drive. This phase inversion reverses the coherent phase accumulation at resonance, suppressing on-resonant excitation while maintaining an off-resonant spectral response across the frequency domain, and thus making resonance distinguishable.

We define the root-echo-Lorentzian $L_{\text{echo}}^{(1/2)}$ drive envelope base on ordered Lorentzian $L^{(1/2)}$ as

$$L^{(n)}(t) = \frac{1}{\left[1 + \left(\frac{t}{\tau}\right)^2\right]^n}, \quad (1)$$

$$L_{\text{echo}}^{(1/2)}(t) := L^{(1/2)}(t) \text{sgn}(t). \quad (2)$$

where τ sets the characteristic pulse duration and n controls the temporal sharpness.

Under the adiabatic condition for the plain Lorentzian pulse, the characteristic spectral width obeys the scaling

$$\Delta_{1/2}\tau \propto (\Omega_0\tau)^{-\frac{1}{2n-1}}, \quad (3)$$

where $\Delta_{1/2}$ denotes the typical linewidth and Ω_0 is the peak Rabi frequency. This relation indicates that in-

creasing the drive amplitude suppresses the detuning at which adiabaticity breaks down, resulting in a narrowing of the effective spectral response. In contrast, the square pulse is fully governed by conventional power broadening, for which the linewidth increases approximately linearly with the drive amplitude $\Delta_{1/2} \propto \Omega_0$. This contrast highlights that the spectroscopic response is not determined solely by the drive strength, but depends sensitively on the temporal envelope and finite-duration boundary behavior of the control pulse, motivating the use of tailored pulse shapes [19–27]. Numerical spectroscopy maps for matched root-Lorentzian and echo-root-Lorentzian pulses are shown at broad and narrow detuning scales in Fig. 2. All three rows use $L = 20 \mu\text{s}$; the Lorentzian-derived pulses use $c = 0.002$, and the columns span 50 and 1 MHz, respectively, with matched model parameters. Independent measurements on qubit q1 on 2026-06-28 give $T_1 = 51.24 \pm 0.73 \mu\text{s}$ and a Ramsey time $T_2^* = 7.31 \pm 0.13 \mu\text{s}$ (Supplemental coherence-characterization section). No Hahn-echo decay was acquired for this data set, so the Letter does not assign the Ramsey result to a measured Hahn-echo T_2 . For a conservative measurement-linked transverse reference, we set $T_{2,\text{eff}} = T_2^* = 7.31 \mu\text{s}$, corresponding to $1/(\pi T_{2,\text{eff}}) = 43.5 \text{ kHz}$. The calculation in Fig. 2 uses these measurement-linked coherence parameters; $T_{2,\text{eff}}$ is not presented as independent Hahn-echo evidence.

Square pulses exhibit pronounced power broadening even at low drive amplitudes compared to Lorentzian-type pulses, with substantial linewidth expansion observed over three orders of magnitude in amplitude. In contrast, the root-Lorentzian pulse displays the characteristic power-narrowing behavior and maintains a narrow linewidth. However, for pulse durations comparable to the qubit lifetime T_1 , coherent oscillations arise

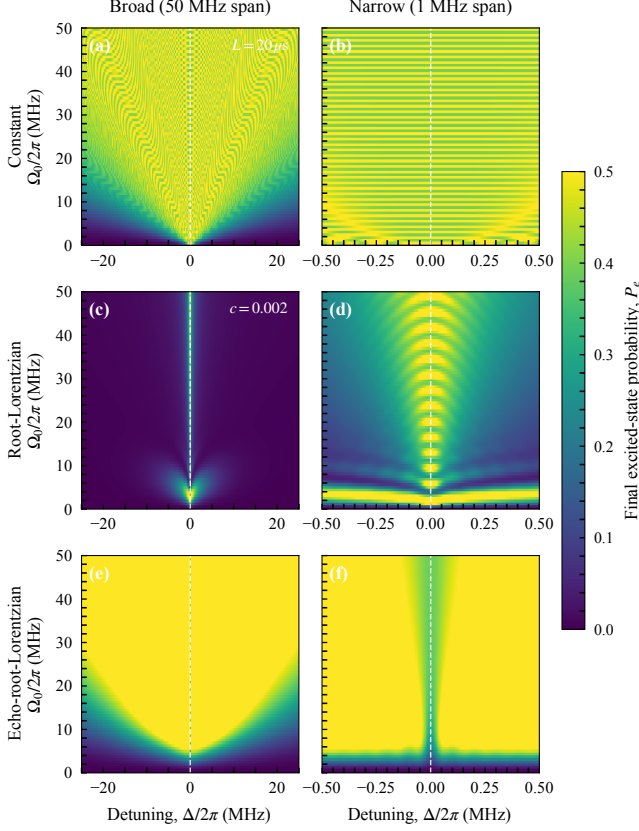


Figure 2. Numerical spectroscopy at two frequency scales for $20 \mu\text{s}$ pulses. The rows show the constant, root-Lorentzian, and echo-root-Lorentzian protocols from top to bottom; the shaped pulses use relative endpoint amplitude $c = 0.002$. The left column spans 50 MHz (-25 to $+25$ MHz), while the right column spans 1 MHz (-0.5 to $+0.5$ MHz). All six panels use the same coherence parameters, peak-Rabi grid, and probability scale. The comparison exposes conventional power broadening for the constant pulse, amplitude-dependent coherent structure for the ordinary root-Lorentzian pulse, and the narrow resonance-centered depletion feature preserved by the echo phase inversion.

for both square and root-Lorentzian pulses, resulting in amplitude-dependent linewidth variations that hinder precise spectral extraction.

Introducing a π phase inversion at the peak of the root-Lorentzian envelope partitions the drive into two sequential segments of opposite phase, such that the corresponding forward and backward evolutions cancel exactly at zero detuning. This echo-like structure suppresses resonant Rabi oscillations while preserving the off-resonant spectral response, producing a sharp depletion feature centered at the qubit transition frequency. In the frequency domain (Fig. 1), this manifests as a splitting of the spectral response accompanied by a null at resonance, rendering the measured linewidth largely insensitive to drive amplitude. This mechanism can be viewed as a Loschmidt-echo-like evolution, in which re-

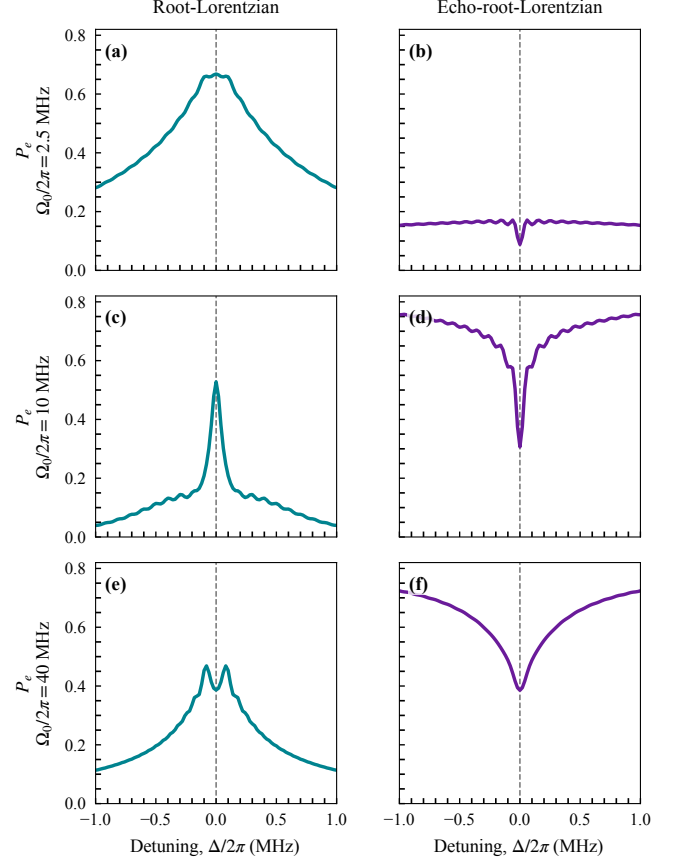


Figure 3. Simulated final excited-state probability P_e versus detuning for $20 \mu\text{s}$ root-Lorentzian (left) and echo-root-Lorentzian (right) pulses with $c = 0.002$. From top to bottom, the matched peak Rabi frequencies are $\Omega_0/2\pi = 2.5$, 10, and 40 MHz. The ordinary root-Lorentzian spectra develop amplitude-dependent coherent structure, whereas the echo spectra retain a resonance-centered depletion feature over the same drive range. All panels use identical coherence parameters, detuning grids, and vertical scales; dashed lines mark zero detuning.

versibility is achieved only on resonance, while detuning or decoherence breaks the cancellation and restores excitation [28, 29]. Combined with the adiabatic following ensured by the Lorentzian envelope, this interference effect suppresses both power broadening and coherent Rabi oscillations, enabling narrow spectroscopy over a broad range of drive strengths and with pulse durations substantially shorter than those required by standard spectroscopy or Ramsey interferometry. Numerical state-evolution maps illustrating the resonant cancellation and detuned response are presented in the Supplemental Material [30].

Figure 3 resolves the two-dimensional maps into fixed-amplitude spectra. The three rows compare the same peak Rabi frequencies for the ordinary and echo protocols, making the echo-induced central depletion and its

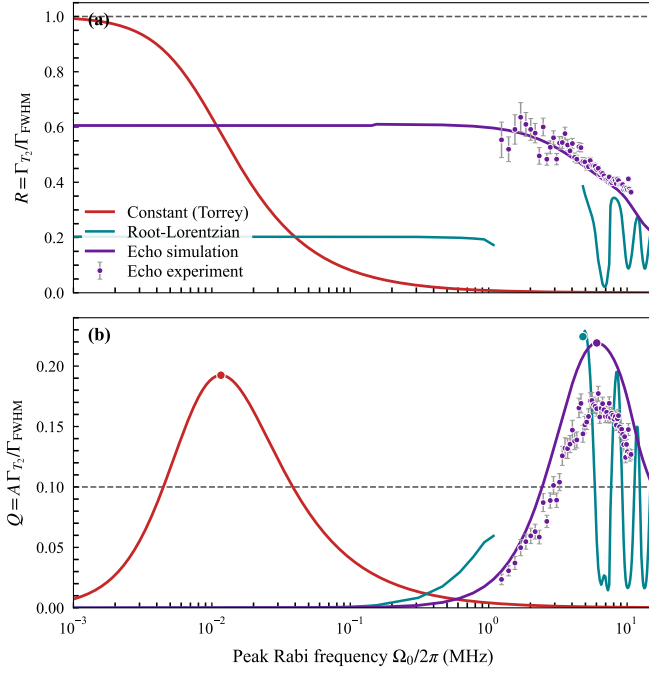


Figure 4. Resolution-contrast comparison for constant, root-Lorentzian, and echo-root-Lorentzian spectroscopy versus peak Rabi frequency Ω_0 . (a) Normalized resolution $R = \Gamma_{T_2}/\Gamma_{\text{FWHM}}$, where $\Gamma_{T_2} = 1/(\pi T_{2,\text{eff}})$. The red curve is the analytic steady-state Torrey result for a constant drive. The teal curve is the ordinary root-Lorentzian simulation, while purple markers are the measured echo-root-Lorentzian values and the same-colored curve is its parameter-matched two-level Lindblad simulation. Both shaped-pulse models use the recorded $10 \mu\text{s}$ duration and $c = 0.005$. The dashed line marks the T_2 -limited reference $R = 1$. Additional model-only points extend the simulations below the first nonzero experimental amplitude, and finite model fits continue above the accepted experimental-fit range; the experimental quality gates are unchanged. (b) Contrast-weighted resolution $Q = AR$. Here A is the fitted peak contrast for the ordinary root-Lorentzian pulse, the fitted dip contrast for the echo protocol, and the steady-state population contrast for the constant drive. The dashed line marks the operational threshold $Q = 0.10$; circular curve markers identify each protocol's maximum accepted Q . Experimental error bars are propagated from the local fit covariance, including the contrast-linewidth covariance in panel (b).

persistence with increasing drive amplitude explicit.

In Fig. 4, we plot the reciprocal linewidth as the normalized resolution $R = \Gamma_{T_2}/\Gamma_{\text{FWHM}}$. The linewidth of the measured echo response is extracted from Gaussian fits. Although the spectral lineshape of a driven two-level system is often expected to follow a Lorentzian profile, the interference-based structure of the present protocol produces a more complex response. In this regime, Gaussian fitting provides a robust and consistent linewidth estimate within our resolution requirements.

We compare the measured echo-root-Lorentzian metrics from the two-dimensional frequency-amplitude

sweep with parameter-matched simulations of both the ordinary and echo root-Lorentzian pulses. The constant-drive reference is the analytic steady-state Torrey solution evaluated using the same measurement-linked T_1 and effective T_2 . Panel (b) reports $Q = AR$, which penalizes a narrow response when its contrast is too small. The three curves therefore expose the useful amplitude window of each protocol directly, including the oscillatory operating structure of the ordinary root-Lorentzian pulse and the broader high-amplitude window retained by the echo protocol.

To suppress residual power broadening at large drive amplitudes, the pulse envelope must satisfy requirements on its cutoff behavior. Truncation introduces nonanalytic boundary behavior and a residual cutoff-broadening contribution that scales with the instantaneous amplitude at the cutoff point, denoted Ω_c [9, 25, 27]. In the model, the condition $T_2^{-1} \leq \Delta_{\frac{1}{2}}$ constrains the maximum allowable cutoff amplitude to $\Omega_c < T_2^{-1}$. Empirically, we find that a relative cutoff below 10^{-3} suppresses excess cutoff broadening even under strong driving conditions; see the Supplemental Material for the cutoff analysis and high-amplitude measurements [30].

At the core of assessing pulse quality and identifying optimal operating regimes lies the spectral response of the driven qubit. We find that the echo pulse yields spectroscopic features with markedly reduced linewidths over a well-defined region of parameter space. We therefore choose pulse durations shorter than the qubit lifetime T_1 but long enough that Fourier broadening does not dominate.

This constitutes an additional advantage of the echo protocol. While both constant and simple Lorentzian pulses exhibit coherent Rabi oscillations that persist even for pulse durations comparable to or exceeding T_1 , the intrinsic π -phase inversion of the echo pulse suppresses these coherence-induced oscillations. As a result, the spectral response remains smooth and well-behaved, enabling reliable linewidth extraction in regimes where standard and Lorentzian pulses display pronounced oscillatory distortions; see the additional high-amplitude data in the Supplemental Material [30].

Second, the use of formally infinite-support pulses necessitates the introduction of a cutoff parameter, as discussed previously. Consequently, the relevant parameter space is effectively three-dimensional: pulse duration, drive amplitude, and cutoff level.

Motivated by the stability observed at fixed pulse duration, we restrict the analysis to constant-length pulses and examine the remaining two parameters. In Fig. 5 we simulate both the root-Lorentzian pulse and its echo-modified counterpart. We extract the full width at half maximum (FWHM) via fitting and evaluate the normalized metric given by the linewidth divided by the peak signal amplitude. We use this quantity as a heuristic

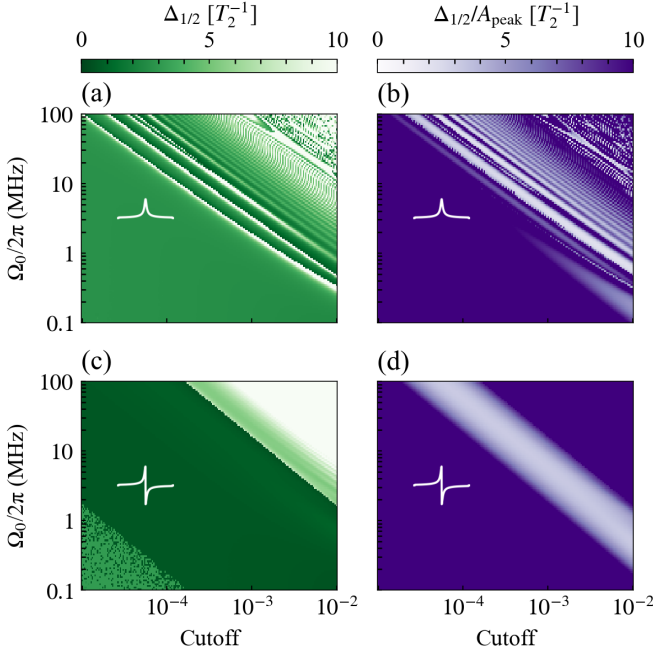


Figure 5. Numerical simulation of spectroscopy scans using root-Lorentzian $L^{1/2}$ and echo-root-Lorentzian $L_{\text{echo}}^{(1/2)}$ drives with a pulse duration of 50 μs . (a),(b) Line width $\Delta_{1/2}$ and the linewidth-to-signal ratio $\Delta_{1/2}/A_{\text{peak}}$ obtained using the root-Lorentzian pulse $L^{(1/2)}$ as a function of peak Rabi frequency Ω_0 and temporal cutoff. (c),(d) Corresponding results for the echo-root-Lorentzian pulse $L_{\text{echo}}^{(1/2)}$. While $L^{(1/2)}$ pulse exhibits extended regions of narrow linewidth, the associated linewidth-to-signal ratio displays strong oscillatory behavior, making it difficult to identify robust operating parameters for high-resolution spectroscopy. In contrast, the echo variant $L_{\text{echo}}^{(1/2)}$ yields both reduced linewidth and a broad parameter region in which the resolution-signal trade-off remains favorable, enabling stable narrowband spectroscopy.

tic resolution-signal metric; a direct Fisher-information or Cramér-Rao analysis would additionally require the measurement-noise model and the full frequency dependence of the response.

The results reveal that the root Lorentzian pulse achieves narrow linewidths across a relatively large region of parameter space. However, its performance is not uniform and exhibits oscillatory structure for the chosen pulse duration. In contrast, the echo-root Lorentzian pulse produces even narrower linewidths while maintaining a significantly broader and more uniform high-resolution region. Although the optimal region may appear small at first glance, the logarithmic scale of the parameter axes should be noted. More importantly, the high-resolution region forms a robust and contiguous “island,” indicating strong parameter insensitivity. This robustness makes the echo protocol substantially more stable against parameter variations and therefore highly advantageous for practical spectroscopy. In conclusion, we have demonstrated that the echo-Lorentzian pulse

constitutes a promising and practical approach for high-resolution spectroscopy. By combining adiabatic shaping with an intrinsic echo phase inversion, the protocol significantly enhances robustness against drive-amplitude variations. Remarkably, it broadens the operational amplitude window by approximately three orders of magnitude, while maintaining high spectral resolution even at drive strengths approaching neighboring qubit transitions. These properties make the echo-Lorentzian protocol particularly attractive for precision qubit characterization under realistic experimental constraints.

This opens the route to coherent spectroscopy based on phase-sensitive off-resonant response rather than population transfer.

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